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FRED SMEIJERS

THE AWARD-WINNING DESIGNER OF FF QUADRAAT AND FONTS FOR OURTYPE, AND AUTHOR OF “COUNTERPUNCH,” DESCRIBES THE TROUBLE WITH “LEGIBILITY” – NOT JUST IN ACHIEVING IT AS A GOAL, BUT DESCRIBING IT AS A CONCEPT.

It is possible, for anyone and certainly for a typedesigner, to experience aha! moments when it concerns legibility. It has happened to me many times and it still goes on. My legibility instinct became very strong and, I am afraid, it will never go away. This is a real professional deformation which, in real life, means that I can hardly look at any text without first shaping an opinion about the typeface/letters in use, the general layout, quality of print, etc. It usually takes at least a few seconds to scan all these before I can start to read. It is quite a relief to realize this acquired reflex can be suppressed in critical situations, like looking for the right check-in gate in a hurry.

A type designer often has to confront himself with the legibility problem on various levels. Onscreen representation of typefaces, for example, is a field where you will often face it in rather unforgiving circumstances. In the mid-'80s I had to experiment with the then so-called “soft-fonts” – screen fonts with greyscaling. The greyscaling effect, when applied well, delivered such a striking legibility improvement that it simply made me feel very, very happy and content. I was whistling while riding back home from work.

Another example, maybe less obvious than the one just given, but also very important to me, was during the development of the Arnhem typeface. Originally Arnhem was a custom-made type design for Staatscourant – a newspaper of the Dutch Government. During the design process, I was in the lucky circumstance to have the opportunity to test in print as many trial fonts as I thought were necessary. We are talking about real conventional legibility here, meaning newspaper print, economy, small sizes while maintaining character and beauty. During testing, one of the things I discovered was that the simple traditional triangular-shaped top serifs functioned better than all the other subtle variations I could come up with. This is, of course, no golden rule. If it is, it only counts for a certain class of conventional typefaces and even then, only when setting large amounts of text in small point size.

So, there we go, talking about legibility means in the first place that you have to be extremely specific about the circumstances in which the reading takes place. The “triangular-serif story” might contain some truth, but it is not necessarily true when it comes to a billboard or when just two words are set in small type size.

The circumstances in which people can and do read nowadays can be so diverse that I have hardly any hope for the general legibility tests and their conclusions, as we knew them from a few decades ago. Reading habits do change and they change, I am afraid, faster than someone like Stanley Morison made us once believe. People adapt and accept a lot when it comes down to reading – or better said: scanning – all kinds of messages.

Of course, there are still many of down to earth do's and don't's within typography. These little laws and rules will often hold ground. But that is not legibility, per se – that is general design knowledge which, in principle, has to lead to better and more readable results. So, in the end, what does legibility exactly mean today for general graphic design? I find that very hard to tell.

On the other hand, I am almost sure that certain products which deal with information transfer are being developed in lab-like circumstances. These products are also being tested for legibility. But legibility there means in the first place not just the letters, but ease of use in general. Layout and user interface structure are in such cases often more decisive than a pixel more or less. – Fred Smeijers

Fred Smeijers is a type designer who specializes in typographic research and development for product manufacturers. Among his typeface designs are FF Quadraat and Quadraat Sans, TEFF Renard, and the OurType Arnhem, Fresco and Sansa. Born in the Netherlands, Smeijers studied graphic design at the Academy of Art in Arnhem. His first practice came in the mid-1980s with the firm of Océ, just then entering the field of typography with laser printers. This set the pattern for Smeijers's long engagement with type design in its most functional applications, as part of product design. After five years he left to work in graphic design, helping to establish the group Quadraat (in Arnhem). The name of the design group was also given to his first published typeface: FF Quadraat, launched by FontShop International in 1992. His work of the 1990s included the expansion of the Quadraat family, type and lettering design jobs for Philips, collaboration on Martin Majoor's Telefont type design, typefaces such as TEFF Renard and Romanee, and his first book “Counterpunch.” With the award of the Gerrit Noordzij Prize in 2000, Smeijers's achievements in the field of practice, research, and education were formally recognized. This prize included a retrospective exhibition of his work, held in The Hague in 2003. His book “Type now” was also published then, as part of the award. That year also saw the launch of the label he co-runs: OurType. In 2004, Smeijers was appointed Professor of Digital Typography at the Hochschule für Grafik und Buchkunst, Leipzig.

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FROGGY SMALL CAPS:

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A MAN OF LETTERS

IF YOU LIKE WHAT YOU READ, YOU SHOULD
THANK MATTHEW CARTER

By Ulrich Boser

Matthew Carter doesn't want you to notice the words you're reading. That is, you shouldn't be aware of the way the small, horizontal line at the top of the h hovers over the T at the beginning of this sentence. Nor should your eye catch on the heavy down strokes of a W that give the letter its classic look. "If the reader is conscious of the type, it's almost always a problem," Carter says. Letters on a page should "provide a seamless passage of the author's thoughts into the reader's minds with as much sympathy, style, and congeniality as possible."

And why should you listen to Carter? Because he designed these very letters you're reading right now—plus dozens of other fonts that appear everywhere from *Sports Illustrated* and the *Pennyroyal Caxton Bible* to muffin-mix packaging and the white pages of the Verizon phone book. His revision of some of the New York Times headline typefaces may soon debut; meanwhile, *businessWeek* will release its redesign on September 26, featuring three fonts fashioned by Carter, two of them custom-made for the magazine.

It's with good reason, then, that Carter has been hailed as the world's most well-read man. At 65, he is the elder statesman of type design, as well as one of its most skilled technical innovators. "He has a phenomenal sense of both history and technology," says Peggy Re, an associate professor of visual arts at the University of Maryland-Baltimore County and curator of the exhibit "Typographically Speaking: The Art of Matthew Carter," a traveling show that opens this week at the University of Pennsylvania. "He will be considered one of the foremost type designers of the 20th century, if not also the 21st century."

A life in type. Born in London, Carter began his career making type the 15th-century way. His father, a famous English type historian and book designer, landed his 17-year-old son an internship at a Dutch type foundry. There Carter cut metal punches much as Gutenberg once did. In 1955, he was accepted to Oxford University—but opted to stay with type instead.

Since then, the arc of his career has followed that of the modern type revolution. In 1966, working for a New York-based type foundry, he created Snell Roundhand, one of the first faces to connect script letters. And in 1981 Carter cofounded the first digital type company, Bitstream, where he designed an early font for laser printers. Later, in 1996, he would develop Verdana, one of the most widely used fonts on the Internet.

And yet some of his most famous works have looked back, not forward. Carter's *Galliard*, which you'll probably find in many of the books on your shelf, was based on a 16th-century font. "Eighty percent of type design is unaffected by technology," he explains. "Letters are a straitjacket, really. You can't on a whim redesign a b so that it ceases to be a b." This tension between the functional and the aesthetic continues to keep Carter, one of 20 or so full-time type designers in the United States, planted in front of a computer screen for hours at a time. "You always have to find some variant which will cause a typeface to be different," he says.

Carter's work cannot be said to have a recognizable style. "Matthew is the quintessential craftsman," says Steven Heller, a design critic and graphic artist. "He is not an artist or experimentalist. He is interpreting conventional forms, and his genius is to take the classical, the traditions of typography, and bring them into the 21st century without seeming trendy. He has a knack for continuing the continuum." While the essence of individual letters hasn't changed for centuries, the digital age has made it far easier to make—and use—new type designs. In the 1950s, there were only a few hundred fonts in the Latin alphabet. Today there are more than 40,000. Microsoft Office XP alone comes with 170 standard fonts. "I used to be afraid of people asking me at dinner parties what I do for a living," Carter says. "Now it amazes me that I can have a perfectly intelligent conversation about fonts with a 9-year-old."

Of his many works, Carter's design for the phone book may have been the most grueling. In 1974, AT&T asked Carter, who was working for Mergenthaler Linotype at the time, to create the smallest legible type that could be printed on low-grade paper. Carter's creation, *Bell Centennial*, has notches at each right angle to prevent ink blotting. The font also has flat, short curves on the sides of g and s in order to increase the white space in the characters and make them more legible.

Letter by letter. Old books and gravestones as well as letters from the Hindi script Devanagari have all served as inspiration for new designs, and Carter's Cambridge, Mass., office is stuffed full with books. But perhaps the most impressive resource for design inspiration comes from his own memory: He can recall, for example, the alphabet his mother cut from linoleum during World War II to help him learn to read. They were a variation of Gill Sans, he says, and the first letters he thought of as objects.

Carter typically starts a new font by sketching the lowercase letters h and o on his Mac.

Then he creates related straight letters like i from the h and round letters like c from the o, keeping a close eye on their weight and overall form by often printing them out; the resolution on a computer screen isn't high enough to see the smallest details. He works last on more "capricious" characters, like the lowercase g and uppercase Q, whose curlicues allow for more artful flourishes. (Experts typically identify fonts by the traits of these letters.) An entire alphabet can take months of painstaking work, with fractions of a millimeter making the difference between an artful letter and an ugly one. "Watching me work is like watching a refrigerator make ice," Carter says.

But individual letters aren't enough to make a good font. "It's only when three or four letters are set together," he says, "that one can start to compare them: 'Look, your h is too big alongside the o. Or, 'it's too thin' or 'it's falling over.' A letter only has properties relative to the letters around it." One reason that Carter

designed the h to hang over the T in the Miller font that you're reading is to ensure that the uppercase letter doesn't overwhelm the word. "It's purely an aesthetic judgment," he says of the feature.

Some of Carter's most innovative recent work has been playing with the space between letters. The Walker typeface, which Carter created in 1995 for the Walker Art Center in Minneapolis, allows users to modify letters by adding what he calls "snap-on serifs." An E, for instance, can be connected by extending its crossbar to an O. Still, as 21st century as that work may seem, Carter refuses to prognosticate about the next big thing: "I've heard so many people talk about the future of type—and none of it's come true—that I've learned to shut up about it."

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**“Watching me work
is like watching a
refrigerator make ice.”**

MATTHEW CARTER